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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	CodeFirst
Selected Domain	Domain 2 — Orbyte Smart SME Challenge
Selected SDG Track (1–17)	SDG 8 — Decent Work and Economic Growth (Targets 8.2, 8.3); also linked to SDG 12 (Target 12.3)
Project / Solution Name	Cafe Replenishment
Name (Leader)	Fwn Jiarui
Email & Phone (Leader)	1486866989@qq.com / +60 175441714
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Independent cafés — micro and small enterprises (SMEs) — must decide how much to buy before tomorrow's demand is known. Weather, weekdays, holidays and menu mix can change sales, while perishable ingredients arrive in fixed packs. Ordering too much can leave edible inputs such as milk or oat milk unused or expired and tie up cash; ordering too little can cause stockouts, disrupted service and lost sales. The people most affected are owner-operators of one- to three-outlet cafés in the Klang Valley, who usually have no dedicated operations analyst: the same person takes orders, manages staff and decides tomorrow's purchases at closing time. Small café teams may have limited time to combine sales history, recipe quantities, on-hand stock, confirmed deliveries and external context in one repeatable process. This is both an operating-cost and environmental challenge: better replenishment decisions could reduce avoidable surplus without increasing stockouts. UNEP estimated that food services generated 290 million tonnes of food waste in 2022, 28% of food waste across the household, food-service and retail sectors [3]. That global figure confirms the sector-wide issue; it is not a Kuala Lumpur café baseline. Cafe Replenishment targets the planning step. The current demo uses synthetic data and does not yet measure real café waste or demonstrate an environmental reduction.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Cafe Replenishment is a browser-based decision aid for the café closing routine. Project-developed ridge-regression models, fitted to deterministic synthetic histories, estimate next-day portions for espresso, latte and oat latte using sales, calendar and weather features. In the public demo, the wider 20-product menu uses an explicit simulated mix; it is not a trained item-level forecast. Recipe/BOM data converts estimated demand into ingredient use, then the replenishment calculation considers on-hand stock, confirmed inbound quantities, safety stock and supplier pack sizes. A manager can review and adjust suggested quantities, then explicitly

approve a simulated purchase draft. The app does not place or pay for orders and does not send email automatically. The published 56-day backtest uses synthetic data only; accuracy for a real café has not been validated.

The prototype uses a native HTML/CSS/JavaScript browser client, CSV recipe data and browser-local demo state. Weather is supplied by Open-Meteo; the demo includes a verified 2026 Kuala Lumpur holiday calendar. Deployment uses Netlify static hosting, Functions and Blobs. Optional, user-controlled integrations include Gmail draft creation and a visitor-selected Runpod model for an explicitly approved paid review; WhatsApp tests require a separate send confirmation. In Phase 2, we plan to build on workshop material in API integration, data modeling and cloud deployment to validate and harden this workflow.

Within the Orbyte ecosystem, this module would consume the hub's shared business context (sales, stock, supplier records) through its open APIs and return an approved replenishment draft to the same approval and audit trail, so one reusable module could serve many micro and small businesses without bespoke integration. The core journey runs end-to-end in the live demo.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Planned prototype stack: native HTML/CSS/JavaScript browser client; CSV recipe/BOM; project-developed ridge-regression forecast; browser localStorage for public-demo state; Netlify static hosting, Functions and Blobs. External services: Open-Meteo weather API; optional Google Gmail API; optional visitor-owned RunPod Serverless endpoint; optional Meta WhatsApp/360dialog test integrations. The public demo keeps all state in the browser, so no central database is required for the prototype; server-side storage would be introduced only when a café partner supplies real sales and stock records.

Figure 1 shows how these components connect into one architecture.

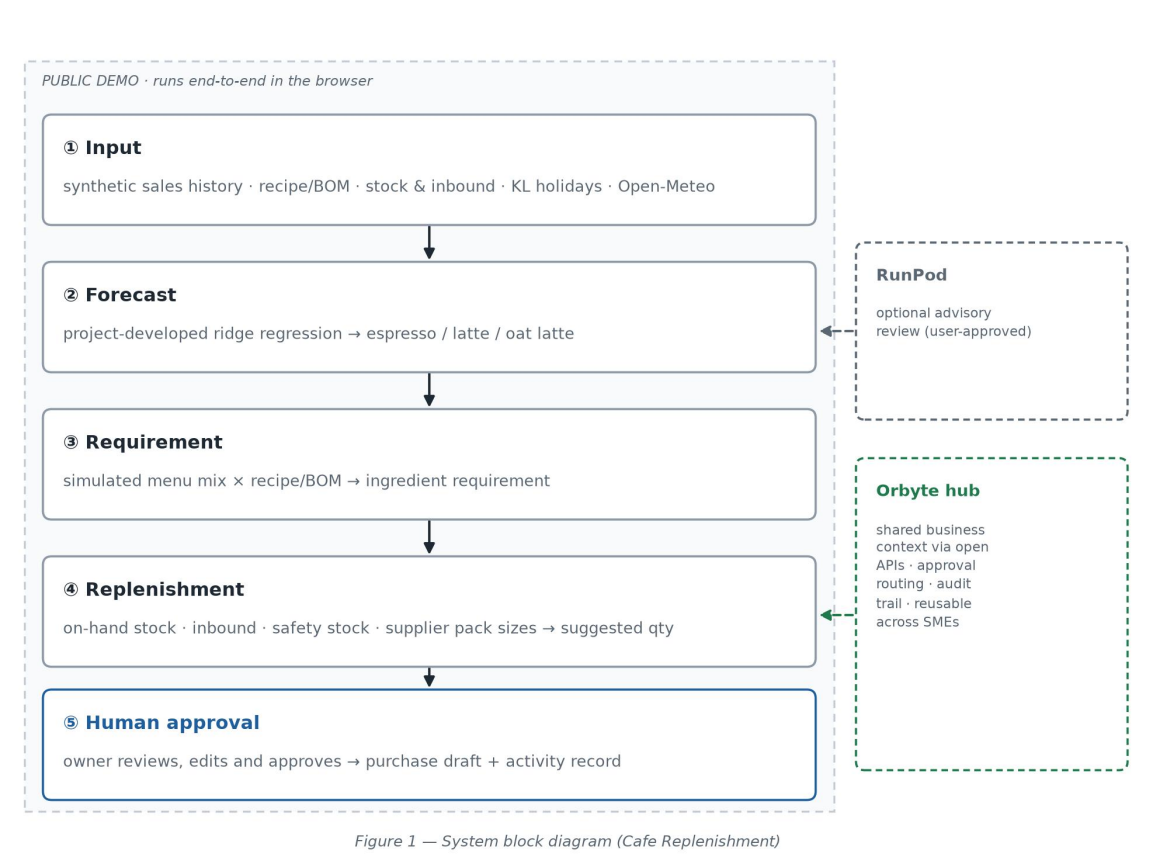


Figure 1 — System block diagram (Cafe Replenishment)

Optional advisory: previewed synthetic context → Netlify Function → visitor's own RunPod model, only after explicit approval. Gmail draft saving and WhatsApp test messages are user-

authorized and require separate confirmation. The Phase 2 plan applies workshop material in API integration, data modelling and cloud deployment.

☑ Yes ☐ No

Replit Core (official SDC-provided, US$20 per participant) was the team's development environment: the browser client, the Netlify Functions and the model-fitting scripts were written, reviewed and debugged inside a Replit Project. The RunPod US$25 team credit covers the optional GPU path, in which a visitor may select an open-source model served from the team's RunPod Serverless endpoint. Neither platform is required for the core replenishment workflow, which runs on Netlify static hosting. Proof of use for both accounts is attached as SDC2026_CodeFirst_Replit_POU.pdf and SDC2026_CodeFirst_Runpod_POU.pdf.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

The current interface follows an owner's end-of-day sequence: check weather and holidays, record sales and stock, review or adjust replenishment, then approve a draft. The independent browser demo labels simulated data and keeps the owner in control of quantities and approval. This makes the workflow technically testable without importing a café's private records. We have no evidence yet of café-owner interviews, usability tests or real-world forecast validation, so those are Phase 2 validation tasks rather than completed results. If a participating café is available, we will observe its closing workflow, agree on a four-week baseline, then pilot the recommendations for four weeks while logging discarded edible ingredients, stockouts and quantity changes. We will interview the owner and revise the flow based on observed friction. Netlify deployment and opt-in integrations provide a path to testing; external payload previews, human approval and separate message confirmation reduce accidental actions. If no café partner is secured, we will report controlled simulations only and make no measured real-world impact claim.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess **Innovation and Creativity** as well as **Feasibility of build**. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary — SDG 8, Decent Work and Economic Growth: Target 8.2 (achieve higher levels of economic productivity through technological upgrading and innovation) and Target 8.3 (promote policies that support the growth of micro, small and medium enterprises) [1]. Secondary — SDG 12, Responsible Consumption and Production: Target 12.3 (halve per-capita food waste at retail and consumer levels by 2030 and reduce food losses along supply chains) [2]. The project contributes operationally to these targets; it does not measure the official SDG indicators.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Public benchmark (global, not café-specific): food services generated an estimated 290 million tonnes of food waste in 2022, about 28% of food waste across households, food services and retail [3]. The project has no real café discard baseline; its demo data are synthetic. Phase 2 baseline: four weeks of weighed edible-ingredient discards and completed orders at a participating café, if secured.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Primary KPI: kilograms of edible ingredients discarded per 100 completed orders. Record discarded edible ingredients by weight and completed orders over the same period; exclude packaging from this KPI. Secondary guardrail: customer-facing stockout rate for tracked ingredients.
Phase 2 Impact Target	Proposed pilot hypothesis: reduce edible-ingredient

Metric Category	Your Response
Your realistic, percentage-based target by the final prototype presentation *e.g., "20% improvement in computational resource use."*	discard kg per 100 orders by at least 10% against the participating café's four-week baseline during a four-week pilot, without increasing stockouts. This is a target, not an achieved result. It depends on obtaining a café partner and recording actual waste; if no partner is secured, report simulation results only and claim no measured real-world reduction.

Sources — [1] United Nations, Sustainable Development Goal 8: Decent Work and Economic Growth, targets 8.2 and 8.3 (https://sdgs.un.org/goals/goal8). [2] United Nations, Sustainable Development Goal 12: Responsible Consumption and Production, target 12.3 (https://sdgs.un.org/goals/goal12). [3] United Nations Environment Programme, Food Waste Index Report 2024 (https://wedocs.unep.org/bitstream/handle/20.500.11822/45230/food_waste_index_report_2024.pdf).

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☒ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☒ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☒ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☒ No

All code, design, research and content in this submission were created by the registered team members during the official hacking period (19–25 October 2026). The public demo is deployed from the team's own repository; its commit history and deployment records are available to the judging panel on request. No pre-existing project, template or prior submission was used as a starting point.

The submission contains no code, design or research that any team member created or owned before 19 October 2026. The single-file prototype that seeded the build (cafe_ollama_stock.py) was written by a registered team member during the hacking period and is preserved unmodified in original/ for traceability; it is credited in NOTICE. Everything submitted as project functionality — the ridge-regression forecast, the recipe/BOM requirement engine, the replenishment and approval workflow, the public browser demo and the Netlify deployment — was created by the registered team members between 19 and 25 October 2026.

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Open-Meteo forecast data/API	Public data / API	Open-Meteo public API terms — free for non-commercial use with attribution
Kuala Lumpur 2026 public-holiday calendar; official	Dataset	Public government sources; attributed in model.json

government sources are listed in the project model metadata		
Optional visitor-selected Runpod-hosted model; not required for baseline demo	Model / external service	Governed by the visitor's chosen provider terms
openpyxl 3.1.5; et_xmlfile (bundled dependency)	Code / libraries	**MIT (both)**

The project uses the third-party data sources, models and libraries listed below. Each is used under its published terms and is attributed in this submission and in the source repository; no third-party asset is presented as the team's own work. Package-level details for the code libraries are declared from the project manifest (package.json).

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☒ No

No team member is bound by an employment, internship or consulting agreement under which an employer could claim ownership of work created during this hackathon.

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Signature: _____ Name: Fwn Jiarui Date: 24 October 2026